Refined observations on enhanced sinusoidal signal detection in extremely noisy time series using R

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Abstract:

This study revisits the challenge of identifying stationary sinusoidal signals submerged in high-noise time series—extending previous work by emphasizing the broader implications, potentialities, and interesting nuances uncovered during our research. Operating within the R programming environment, our methodology tackles noise magnitudes up to 20 times higher than the target signal (SNR as low as 1/20).

In contrast to standard approaches, we employ an integrated set of techniques—spectral differentiation, minimal-window Kolmogorov-Zurbenko filtering, dataset repetition, and zero padding—that consistently outperforms ten well-known CRAN packages under varying levels of noise intensity. Notably, the synergy among these methods proves pivotal: omitting any single component diminishes the effectiveness of the ensemble. Among our findings is the linear offset observed at high-noise conditions, which, although deviating from the exact frequency, remains consistent and thus offers a predictable pattern for further tuning.

We also discuss potential applications beyond strictly stationary signals, highlighting how future adaptations may integrate wavelet-based or adaptive filtering to address more complex, non-stationary time series. Further exploration of optimal parameterization - such as selecting fewer tapers in multitaper methods - reveals intriguing trade-offs between frequency resolution and leakage prevention. Our results underscore the importance of holistic strategies that unite multiple computational steps, ultimately paving the way for more reliable signal detection in data-intensive fields where noise significantly hinders traditional spectral analysis.

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Introduction:

- Detecting weak periodic signals in noisy time series is a long-standing challenge.
- Performance drops sharply when signal-to-noise ratio (SNR) falls below 1/20.
- Traditional spectral methods often fail under extreme noise.
- multi-step strategies can improve detection.
- Focus is on stationary signals, with potential for broader applications.
- Highlights consistent patterns in high-noise environments.
- Aims to support more reliable signal recovery across data-intensive fields.

Method/Materials:

- Applied to synthetic time series with stationary sinusoidal signals.
- Target SNR as low as 1/20 (noise 20× stronger than signal).
- Combined four techniques to enhance signal visibility:
  - Spectral differentiation – sharpens frequency components.
  - Minimal-window KZ filtering.
  - Dataset repetition reinforces periodicity.
  - Zero padding – improves frequency resolution.
- Benchmarked against widely used CRAN spectral analysis tools.
- Analyzed frequency offset patterns at extreme noise levels.

Results:

★ Method consistently outperformed CRAN packages.

★ Signal freq was accurately detected even at SNR = 1/20.

★ Removing any single component reduced accuracy.

★ Observed a linear frequency offset under high noise.

★ Method maintained robustness across different noise level

★ Improved clarity and resolution of spectral peaks

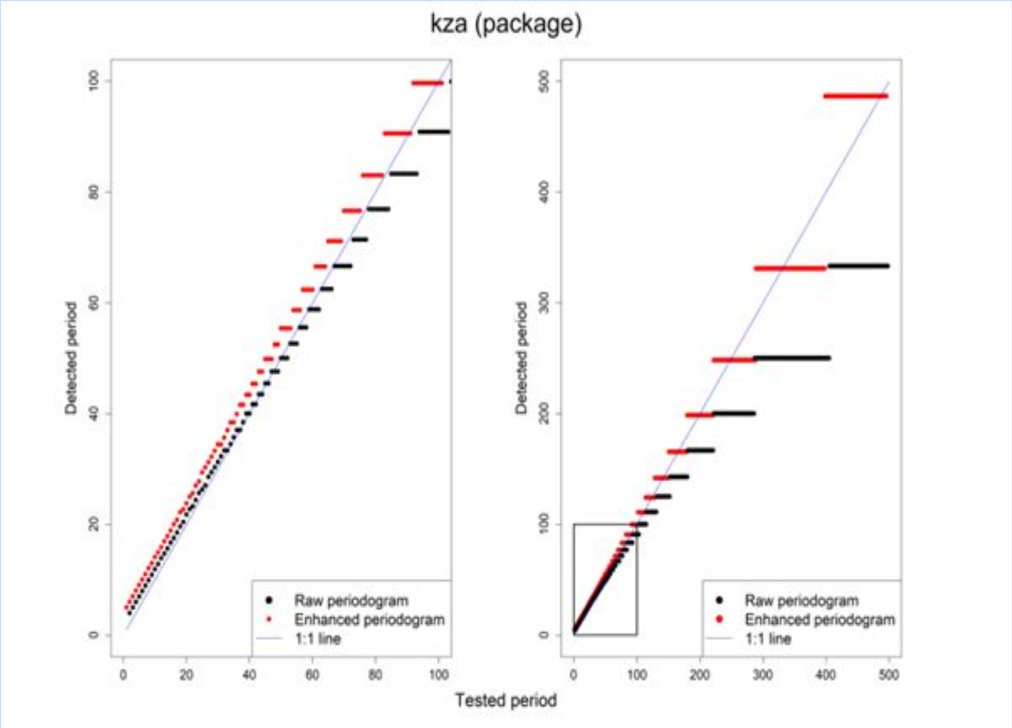
Discussion / Conclusions

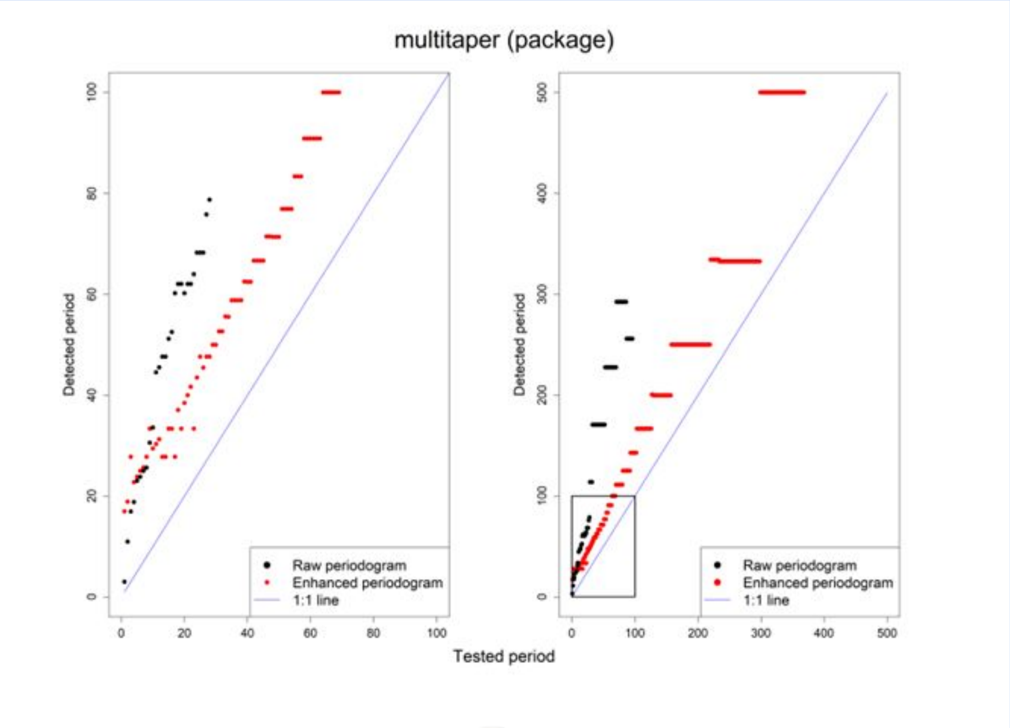
❖ Results highlight the strength of combining multiple spectral techniques.

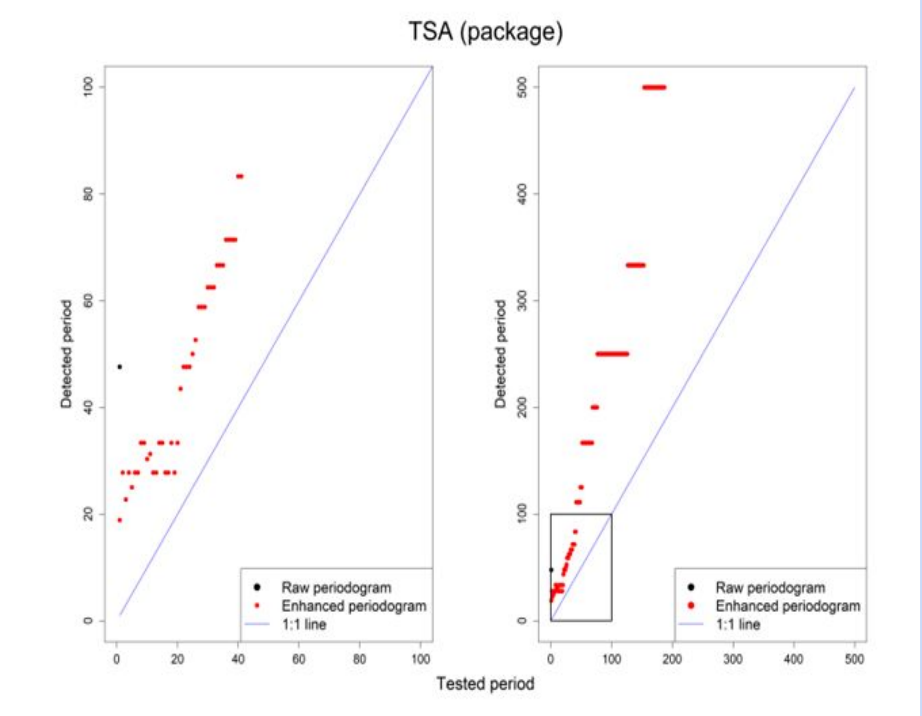
❖ Predictable offset under high noise can aid in post-processing adjustments.

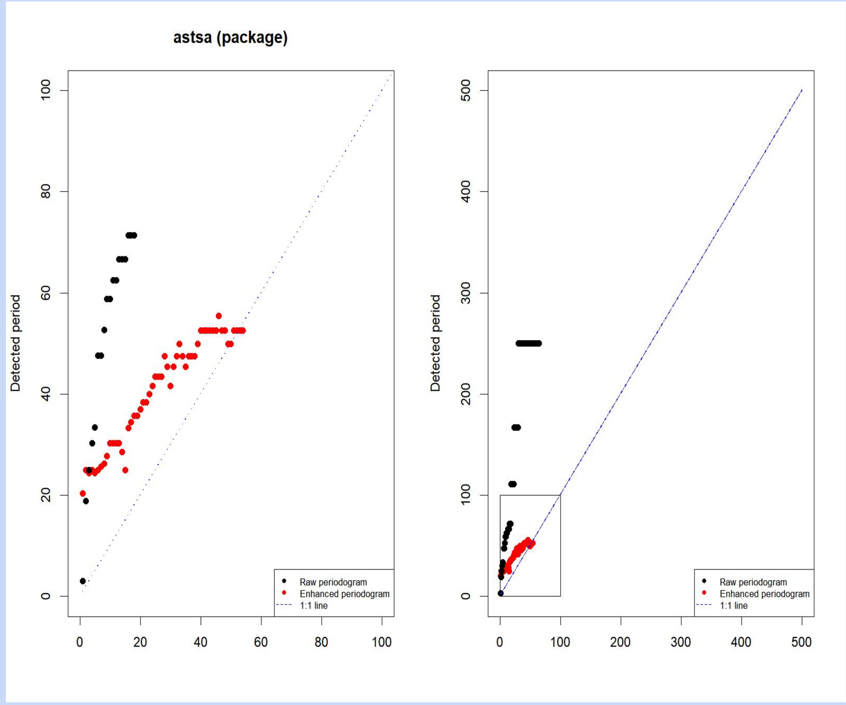
❖ Ensemble approach offers greater reliability than any single method.

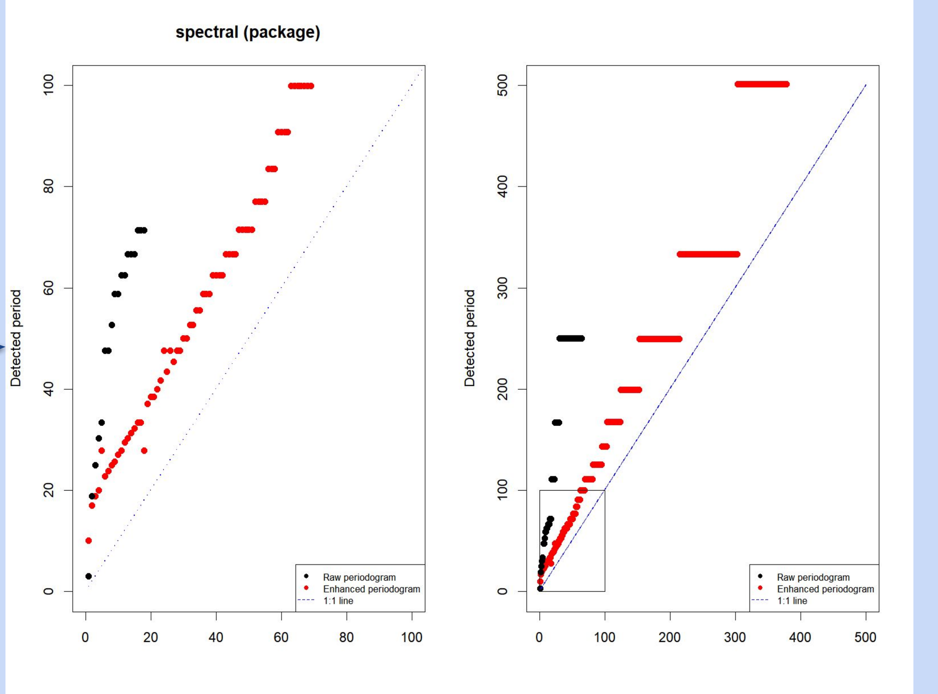
❖ Potential to adapt this strategy for non-stationary signals using wavelets or adaptive filters.

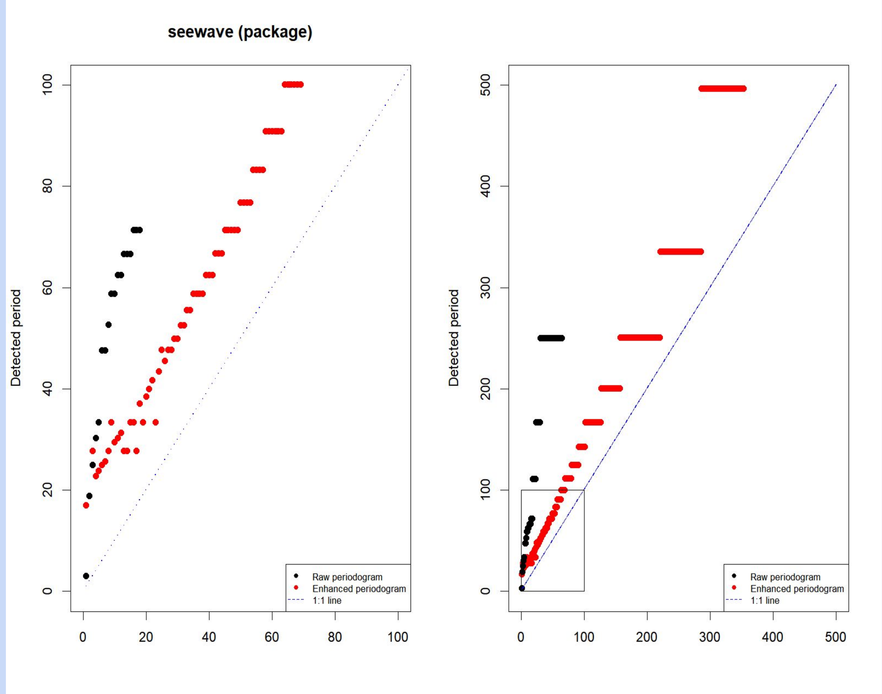




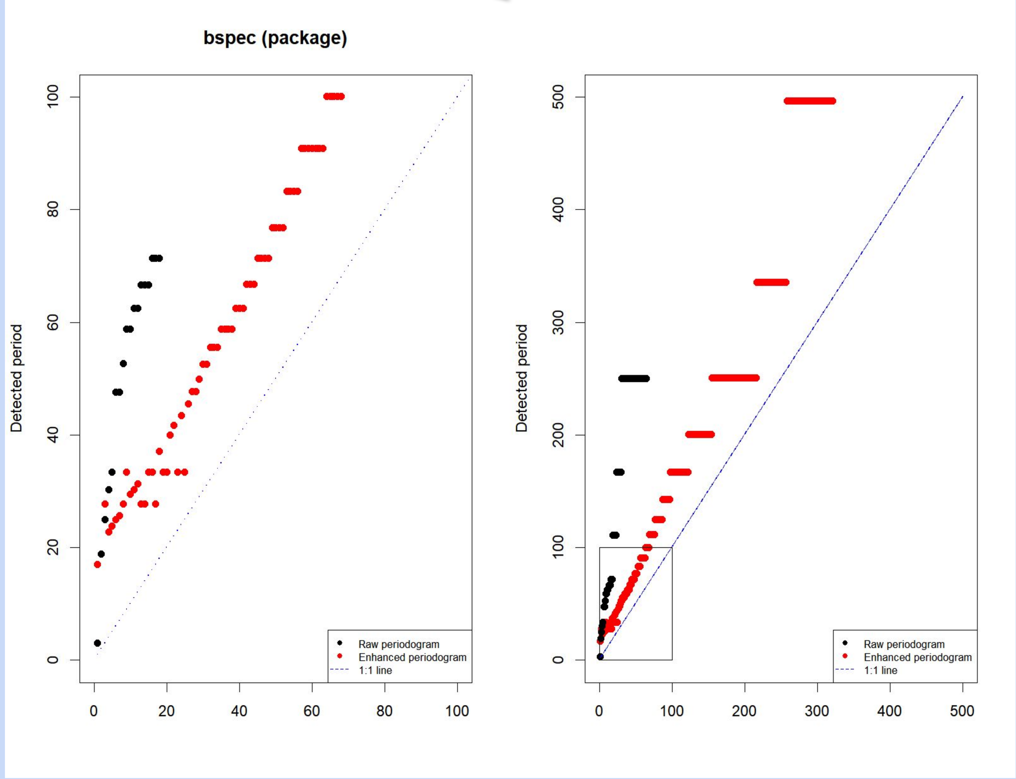


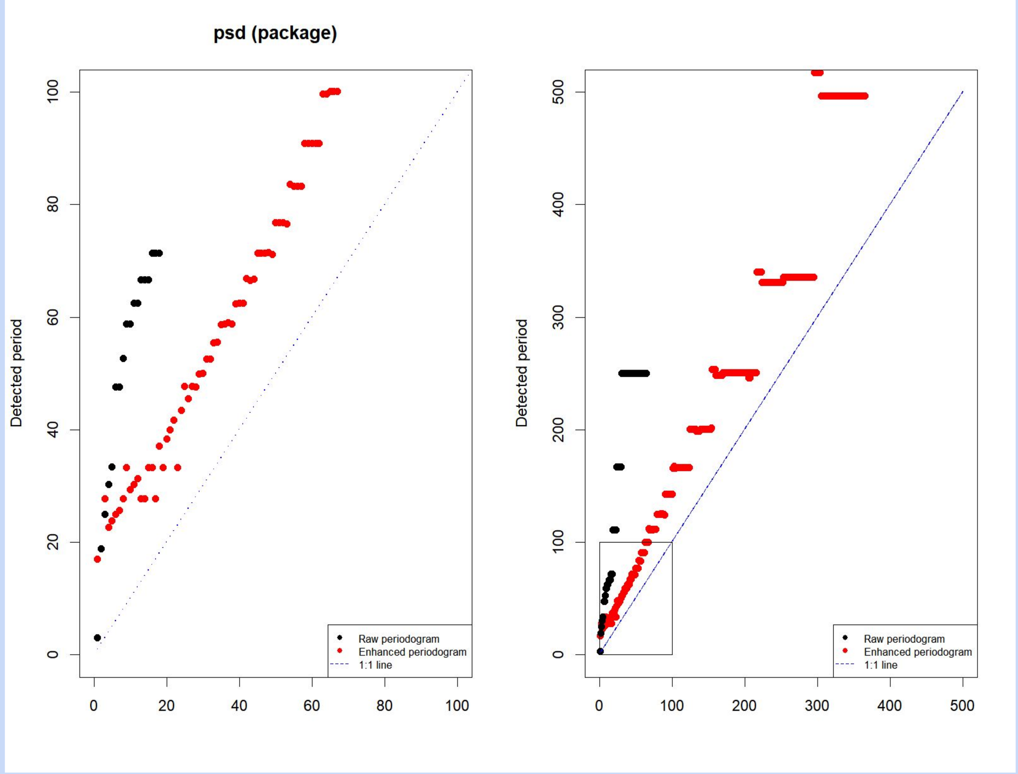






Signal Detection Comparison across many CRAN packages.
Red dots represent frequencies detected using CRAN packages with our enhanced methodology, while black dots show standard multitaper results under 1/10 signal-to-noise ratio. The blue line indicates perfect detection accuracy, with red dots demonstrating closer alignment and better performance.





References:

Barbour, A.J., Parker, R.L. *psd: Adaptive, sine multitaper power spectral density estimation for R*. Computers & Geosciences 63, 1–8 (2014).
Chan, K., Ripley, B. *TSA: Time Series Analysis*. R package version 1.3.1, https://CRAN.R-project.org/package=TSA, (2022).
Cheng, H.L., Han, J.Y., Zheng, W.Z., Cheng, Y.F., Chu, Y.C., Lin, C.M., Chiang, M.C., Liao, W.H., Lai, Y.H. *Objective Signal Analysis for Investigating Feasibility of Active Noise Cancellation in Hearing Screening*. Sensors (Basel) 22(19), 7329 (2022).
Close, B., Zurbenko, I., Sun, M. *kza: Kolmogorov-Zurbenko Adaptive Filters*. R package version 4.1.0.1, https://CRAN.R-project.org/package=kza, (2020).
Damaschke, N., Kühn, V., Nobach, H. *Bias-free estimation of the covariance function and the power spectral density from data with missing samples including extended data gaps*. EURASIP Journal on Advances in Signal Processing 2024, 17 (2024).
Felix, L.B., Sá, A.M.F.L.M., Mendes, E.M.A.M., Moraes, M.F.D. *Post-processing of auditory steady-state responses to correct spectral leakage*. Journal of Neuroscience Methods 181, 145–149 (2009).
Harris, F.J. *On the use of windows for harmonic analysis with the discrete Fourier transform*. Proceedings of the IEEE 66(1), 51–83 (1978).
Jwo, D.J., Chang, W.Y., Wu, I.H. *Windowing techniques, the Welch method for improvement of power spectrum estimation*. Computers, Materials & Continua 67(3), 3983–4003 (2021).
Lessard, C.S. *Window Functions and Spectral Leakage*. In: Signal Processing of Random Physiological Signals. Synthesis Lectures on Biomedical Engineering. Springer, Cham (2006).
Nasser, A., Mansour, A., Yao, K.C., et al. *Spectrum sensing based on cumulative power spectral density*. EURASIP Journal on Advances in Signal Processing 2017, 38 (2017).
Nuttall, A.H. *Some windows with very good sidelobe behavior*. IEEE Transactions on Acoustics, Speech, and Signal Processing 29(1), 84–91 (1981).
Park, K.S., Hong, S., Park, P., et al. *Spectral Content Characterization for Efficient Image Detection Algorithm Design*. EURASIP Journal on Advances in Signal Processing 2007, 082874 (2007).
Prieto, G.A., Parker, R.L., Thomson, D.J., Vernon, F.L., Graham, R.L. *Reducing the bias of multitaper spectrum estimates*. Geophysical Journal International 171(3), 1269–1281 (2007).
Seilmayer, M. *spectral: Common Methods of Spectral Data Analysis*. R package version 2.0, https://CRAN.R-project.org/package=spectral, (2021).
Signal developers. *signal: Signal Processing*. https://r-forge.r-project.org/projects/signal, (2023).
Stoffer, D., Poison, N. *astsa: Applied Statistical Time Series Analysis*. R package version 2.0, https://CRAN.R-project.org/package=astsa, (2023).
Sueur, J., Aubin, T., Simonis, C. *Seewave: a free modular tool for sound analysis and synthesis*. Bioacoustics 18, 213–226 (2008).
Tsakiri, K., Marsellos, A., Kapetanakis, S. *Artificial Neural Network and Multiple Linear Regression for Flood Prediction in Mohawk River, New York*. Water 10, 1158 (2018).
Tsakiri, K.G., Zurbenko, I.G. *Effect of noise in principal component analysis*. J. Stat. Math. 2, 40 (2011a).
Tsakiri, K.G., Zurbenko, I.G. *Prediction of ozone concentrations using atmospheric variables*. Air Qual. Atmos. Health 4, 111–120 (2011b).
Yang, W., Zurbenko, I. *Nonstationarity*. Wiley Interdiscip. Rev. Comput. Stat. 2, 107–115 (2010).